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Thermal conductivity and specific heat of a glass-epoxy composite at temperatures below 4 K

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The thermal conductivity and specific heat of the glass-cloth reinforced epoxy laminate G-10, as well as the cryogenic grade G-10 CR, have been measured over the temperature range 0.1–4 K.

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A glass-cloth reinforced epoxy laminate designated by the National Electrical Manufacturers' Association as NEMA G-10 has been used frequently in cryogenic applications. The reader may be more familiar with this green-colored material in its use for printed-circuit boards. More recently, a cryogenic-grade laminate, G-10CR, has become available.

The purpose of the work reported in this paper was to determine the usefulness of these composites at temperatures below 1 K. To this end, both the thermal conductivities and specific heats have been measured using techniques similar to those described previously.¹ Mechanical properties at temperatures, T , down to 4 K have been reported previously.² These mechanical properties should be independent of temperature for $T < 4$ K.

The thermal conductivities for heat flow parallel to the plane of the glass laminates³ are plotted in Fig. 1. Previous work on G-10 and G-10CR above 2 K is also shown.^{2,4,5} The thermal conductivity perpendicular to

the plane of the glass laminates was not measured here, but it has been reported for G-10CR at $T \geq 2$ K as being $\approx 20\%$ smaller.² An estimate for the thermal conductivity of a composite with the glass-epoxy ratio of G-10 is shown by the dashed line in Fig. 1. This calculation⁶ was based on the measured thermal conductivities for fused quartz⁷ and a particular epoxy.⁸

Care should be taken in extrapolating the thermal conductivity data of Fig. 1 below 0.1 K using the approximately $T^{1.8}$ temperature dependence which is characteristic of both glass and epoxy. This is because phonon scattering at the glass-epoxy interfaces may become increasingly important. With decreasing temperature, the phonon mean free paths in the epoxy⁹ and in the glass⁷ increase. When the mean free path is comparable to the interface separation of $\approx 10^{-5}$ m (the diameter of the glass fibers⁶), scattering off these interfaces due to the difference in acoustic impedance can depress the thermal conductivity. This was calculated to occur at $T \lesssim 0.3$ K. A hint of this behavior can be seen in the present data as the thermal conductivity starts to fall more rapidly than $T^{1.8}$ at the lowest temperatures.

The specific-heat data are presented in Fig. 2. The dashed curve is again an estimate using the measured specific heats of fused silica⁷ and an epoxy.⁹ The present data on G-10 and G-10CR agree well with this estimate at $T \geq 4$ K, but at lower temperatures the specific heats of G-10 and G-10CR are much larger. A large low-temperature specific heat is typical of many commercial glasses and ceramics; this extra heat capacity is attributed to iron impurities. In Pyrex, for example, the iron content is ≈ 0.02 wt%. Since the specific heat of G-10 is a factor of ≈ 5 larger than for Pyrex¹⁰ near 0.5 K, we would expect the iron content of G-10 to be ≈ 0.1 wt%. Indeed, an analysis of G-10 showed 0.21 wt% iron, lending support to our suggestion that iron is the cause of the large specific heat.

The composites are easily machined, though tools are rapidly dulled by the glass fibers. In addition, the composites are readily bonded by the application of epoxies. However, the large specific heats of these composites could be deleterious to low-temperature applications if it is desired that structural components have a rapid thermal-response time. Based on the thermal properties

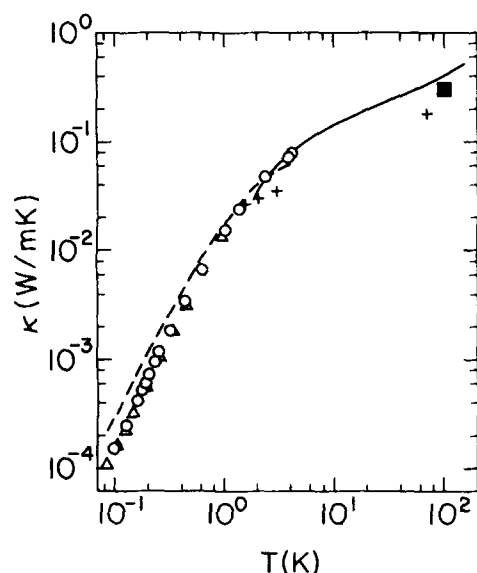


FIG. 1. Thermal conductivity in a direction parallel to the glass fabric, vs temperature. $\circ$, G10; $\triangle$, G10CR; $+$, G10 from Ref. 4; solid line, G-10CR from Ref. 2; $\blacksquare$, average of all data on glass-fiber/epoxy composites from Ref. 5; dashed line, calculated as discussed in text.

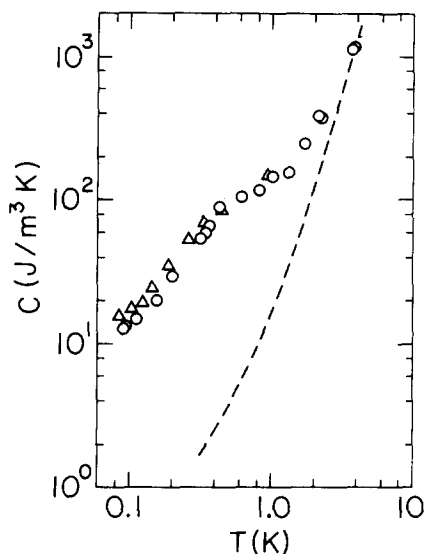


FIG. 2. Specific heat vs temperature. ○, G-10; △, G-10CR; dashed line, calculated as discussed in text.

shown in Figs. 1 and 2, the cryogenic grade G-10CR is not preferable to the readily available G-10 grade.

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- ² M. B. Kasen, G. R. MacDonald, D. H. Beckman, and R. E. Schramm (to be published).
- ³ The G-10 sample was obtained from our stockroom. The sample of G-10CR was manufactured by Micarta Division, Westinghouse Electric, Hampton, North Carolina, and was designated lot 56 G12-AA.
- ⁴ J. A. Barclay, S. S. Rosenblum, and W. A. Steyert, *Cryogenics* **16**, 539 (1976). The figures in this paper are interchanged. The thermal conductivity of G-10 appears in Fig. 8, not Fig. 6.
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- ⁶ Our estimate of the mass density of G-10 is 1740 kg/m³ with ≈42% volume fraction of resin, and of G-10CR is 1780 kg/m³ with ≈38% volume fraction of resin. The glass fibers in the G-10 are ≈10 μm diam.
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Low-frequency tunable (1–1.8 GHz) electron spin resonance resonator and spectrometer

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We describe a tunable low-frequency Electron Spin Resonance spectrometer suitable for operation in the liquid helium temperature range. The resonator is a one-quarter wavelength piece of stripline, whose frequency can be set anywhere in the band from 1–1.8 GHz. The microwave bridge uses a solid-state oscillator and commercially available components.

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Most Electron Spin Resonance (ESR) measurements utilize microwave frequencies at x band (~9 GHz) and above. These frequencies are useful to resolve structure in the ESR spectrum, but there are advantages to investigating the ESR at low frequencies.¹ We describe here a low-frequency ESR spectrometer and resonator which allow the measurement of ESR spectra in a frequency range of 1–2 GHz at helium temperatures. The present instrument differs from previous designs² in that it is particularly compact, and well suited to cryogenic operation.

The spectrometer is assembled from commercial

microwave components. The resonator is a quarter-wavelength stripline conductor whose frequency can be adjusted with a capacitive element while the cavity is at room temperature. As used here, the resonator can be set at any frequency from 1–1.8 GHz, but the range could be extended to 2 GHz with minor modifications. In the next section we will describe the mechanical features of the resonator and its operation, followed by a brief description of the spectrometer bridge and the frequency locking of the spectrometer to the resonator.

Figure 1 is an exploded view of the resonator. The center conductor of the stripline is attached to the ground